

*TESTING*  
**CLIMATE  
STRATEGIES**  
*THROUGH SIMULATION*



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# I. THE EN-ROADS MODEL

## b. System Dynamics

### a. What is it ?

- Global climate simulator
- Developed by Climate Interactive and MIT
- Observe different climate policy levers' effects up to 2100

- Stocks : quantities accumulated over time
- Flows : variations that modify stocks

Fundamental equation :

$$\frac{dS}{dt} = \sum(\text{inflows}) - \sum(\text{outflows})$$

>> system of coupled, non linear differential equations

- Positive loop (amplification) :

temperature ↗ → ice melt ↗ → albedo ↘ → temperature ↗

- Negative loop (stabilisation) :

carbon tax ↗ → fossil fuel prices ↗ → clean investments ↗ → emissions ↘

# c. Numerical Resolution

- Fixed time step : 0.125 years
  - Numerical method : deritive of Euler's method
  - Simulation from 1990 to 2100
- >> several thousand time steps

# d. Key Variables

| Variable                   | Subsystem | Role  | Unit         | Effect if carbon tax ↗ |
|----------------------------|-----------|-------|--------------|------------------------|
| Relative fossil fuel price | Energy    | Aux.  | \$ / MWh     | Increase               |
| Solar / wind share         | Energy    | Aux.  | %            | Increase               |
| Energy emissions           | Climate   | Flow  | GtCO2 / year | Decrease               |
| CO2 concentration          | Climate   | Stock | ppm          | Decrease (delayed)     |
| Fossil capacity            | Energy    | Stock | GW           | Decrease               |
| Fossil production          | Energy    | Flow  | EJ / year    | Decrease               |
| Carbon intensity           | Energy    | Aux.  | tCO2 / EJ    | Decrease               |
| Final energy demand        | Demand    | Flow  | EJ / year    | Decrease               |
| Electrification rate       | Demand    | Aux.  | %            | Increase               |
| Global temperature         | Climate   | Stock | °C           | Decrease (delayed)     |

# II. KEY SUBSYSTEMS

## A) ENERGY

- Where does it act ?
- Main mechanism :
- Type of impact :
- Temporality :
- Limit :

Energy supply and carbon intensity

Investments → capacities → production → emissions

Structural, sustainable, progressive

Slow initially, powerful in the long term

If demand remains high, transition is more costly and slower

>> Economics signals guide investments, which progressively modify stocks and determine emissions.

# II. KEY SUBSYSTEMS

## B) DEMAND & USAGE

- Where does it act ?
- Main mechanism :
- Type of impact :
- Temporality :
- Limit :

On the total quantity of energy consumed

Efficiency → demand ↘

Electrification → Carbon intensity ↘

Direct, rapid effects

Faster than energy, depends on equipment renewal (10-20 years)

If electricity remains carbon intensive, electrification alone is insufficient

Causal chains :

Efficiency ↗ → Demand ↘ → Emissions ↘

Electrification ↗ → Carbon intensity ↘ → Emissions ↘



## II. KEY SUBSYSTEMS

### C) SUMMARY TABLE OF SUBSYSTEMS

| Subsystem              | Acts on                   | Effect Type  | Speed         | Long-term Power   |
|------------------------|---------------------------|--------------|---------------|-------------------|
| Energy                 | Supply / carbon intensity | Structural   | Slow          | Very strong       |
| Demand & Usage         | Energy quantity           | Direct       | Medium / fast | Strong            |
| Land & Land Use        | Carbon absorption         | Compensatory | Slow          | Limited           |
| Climate & Carbon Cycle | CO2 stock / temperature   | Final result | Very slow     | Depends on others |



# III. POLICY PARAMETERS

## A) WHICH LEVERS HAVE THE MOST IMPACT ?

| Lever           | Value        | Start | Assumptions            | Expected Result | Sustainability           | Feasibility           |
|-----------------|--------------|-------|------------------------|-----------------|--------------------------|-----------------------|
| Carbon tax      | 100\$ / t    | 2028  | Household compensation | -0.5°C          | Sustainable (30-40years) | Socially difficult    |
| Efficiency      | +2.5% / year | 2026  | Renovation aid         | -0.4°C          | Sustainable (renewal)    | Slow and costly       |
| Electrification | 25% subsidy  | 2026  | Battery cost reduction | -0.7°C          | Sustainable (10-20years) | Huge budget           |
| Renewables      | Subsidy      | 2027  | Storage progree        | -0.5°C          | Sustainable              | Storage needed        |
| Forests         | 58% of land  | 2027  | Certification          | -0.1°C          | Sustainable but limited  | Physical limt         |
| Methane         | -40%         | 2026  | Practice changes       | -0.2°C          | Limited (short term)     | Agricultural barriers |

# III. POLICY PARAMETERS

## B) CARBON TAX

- Mechanism : Tax  $\nearrow \longrightarrow$  Fossil fuel prices  $\nearrow \longrightarrow$  Clean investments  $\nearrow \longrightarrow$  Emissions  $\searrow$
- Strengths : Systemic, very sustainable
- Weaknesses : Social acceptability
- Example : Sweden ( €130/t since 1991 )

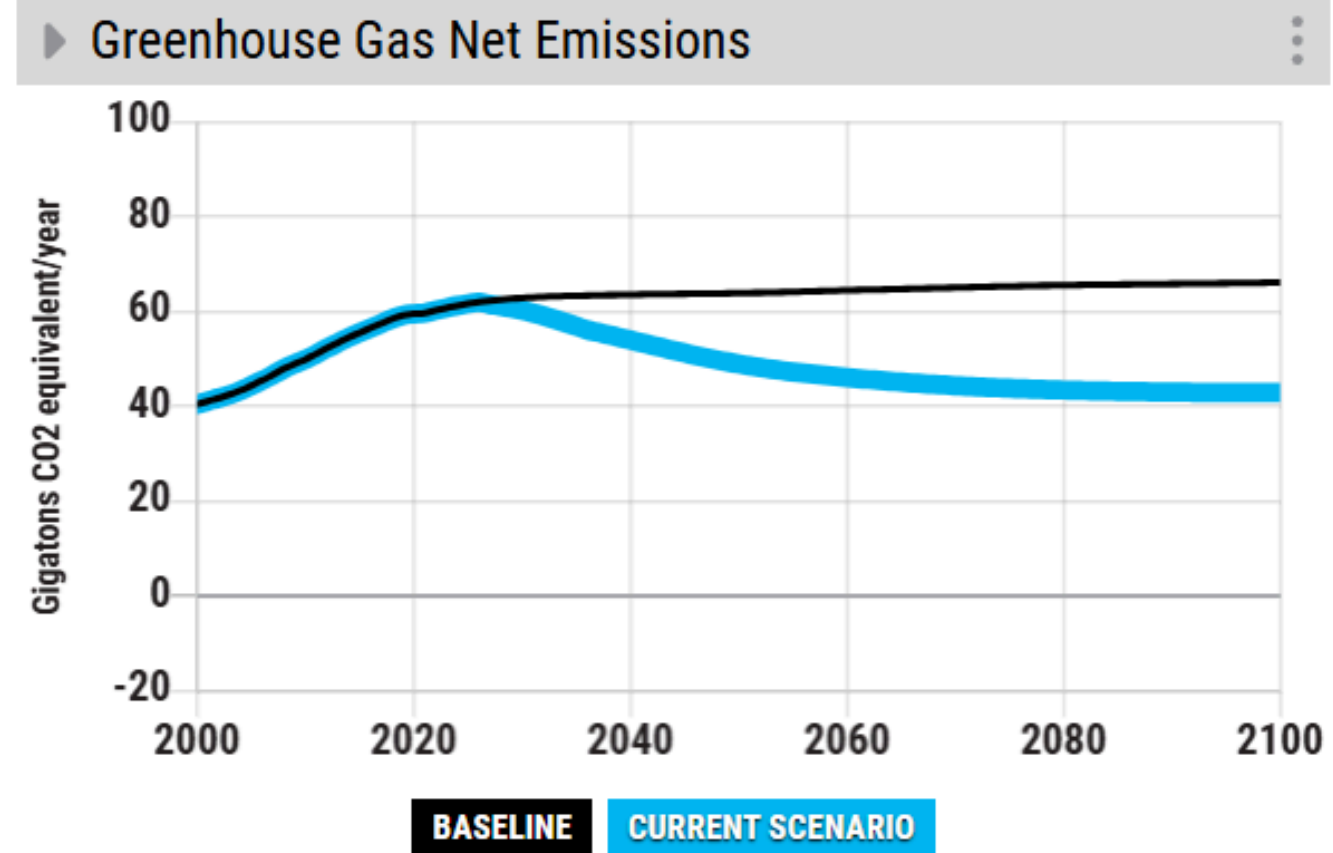
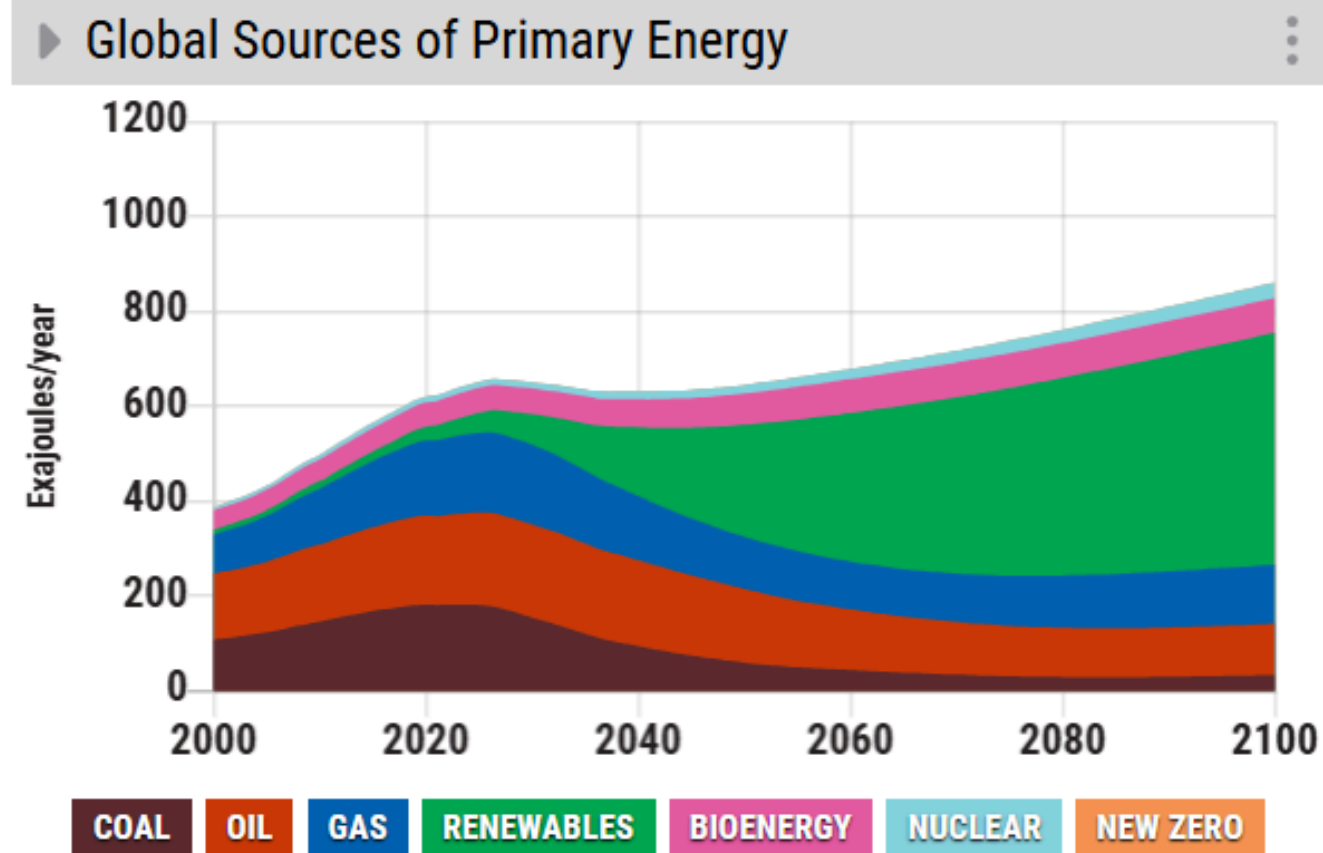
## C) ELECTRIFICATION

- Mechanism : Electric uses  $\nearrow$  + Clean mix  $\longrightarrow$  Carbon intensity  $\searrow \longrightarrow$  Emissions  $\searrow$
- Strengths : Sustainable (10–20 years), acts on usage
- Weaknesses : Budget, grid, decarbonized mix required
- Examples : Norway (80% electric sales), China



# III. POLICY PARAMETERS

## ENERGY SUBSYSTEM



**+2.8°C**

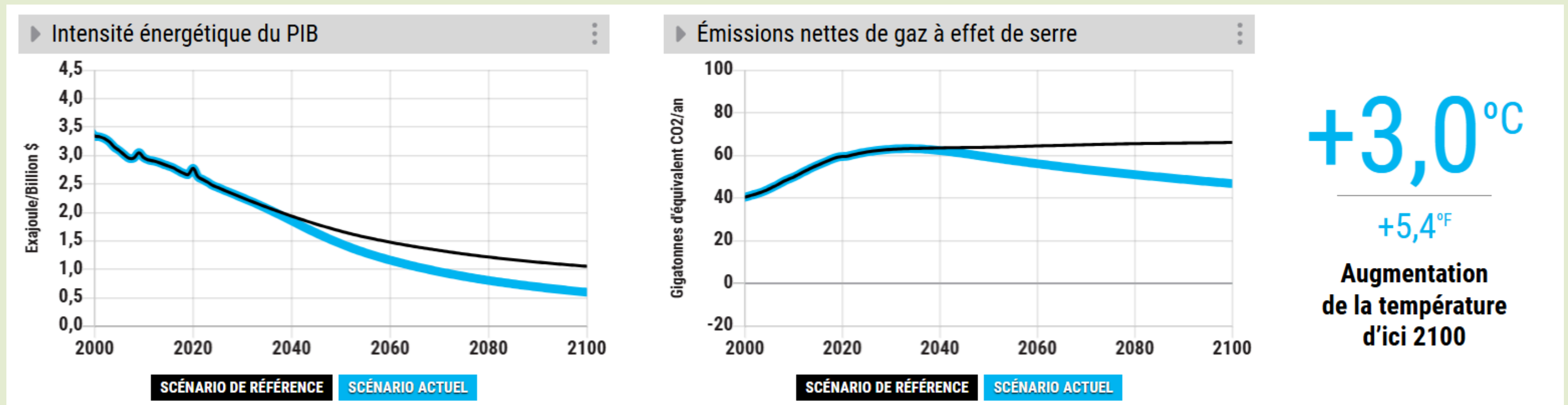
+5.0°F

Temperature Increase by 2100

Carbon Tax + Renewable Energies

# III. POLICY PARAMETERS

## DEMAND & USAGE SUBSYSTEM



Building and Transport Electrification and Energy Efficiency

# III. POLICY PARAMETERS

## D) SYNTHESIS

- Carbon tax and renewables : structurally decarbonize supply
- Efficiency and electrification : reduce demand and transform usage
- Forests and methane reduction : essential supporting roles

# IV. SCENARIOS

## a. Reference scenario

- Carbon tax : non existent
- Energy efficiency : slow progression (1-1.5% / year)
- Electrification : current trends without acceleration
- Renewable energie : moderate development
- Deforestation : continues at current rate
- Methane : no targeted reduction

**RESULT : +3.3°C IN 2100**

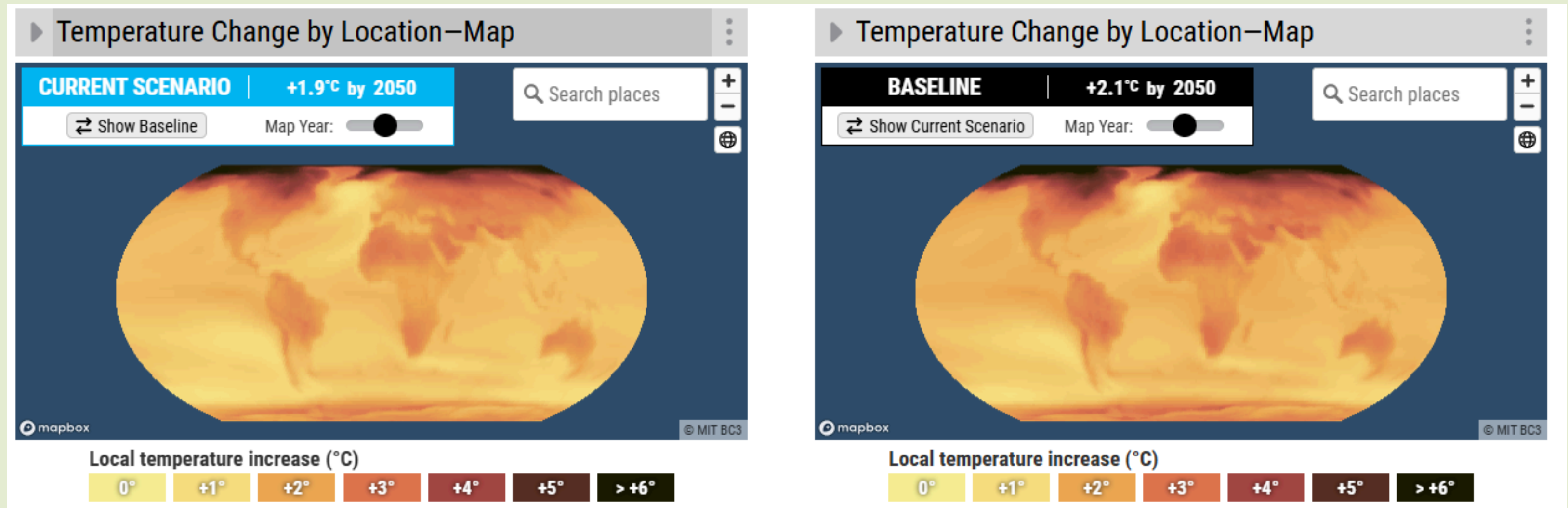
# IV. SCENARIOS

## b. Complete scenario

| Trial | Lever(s)                     | Values                   | Start     | Assumptions                                                  | Recorded Indicators                        |
|-------|------------------------------|--------------------------|-----------|--------------------------------------------------------------|--------------------------------------------|
| 1     | CO2 Tax                      | 100\$/t                  | 2028      | Substitution to rewards<br>renewables                        | Emissions, renewable share,<br>temperature |
| 2     | Efficiency + Electrification | +2.5% / years *          | 2026      | Progressive implentation                                     | Impact on consumption<br>and energy bill   |
| 3     | Methane + other gases        | -40%                     | 2026      | Rapid impact on temperature                                  | 60.52 Gt/CO2 / year in<br>2050             |
| 4     | Nature (Forests)             | +3 Gt (Aff) / -50% (Def) | 2027/2028 | Creation of carbon sinks for<br>"Net Zero"                   | CO2 absorption, land use                   |
| 5     | Complete scenario            | All previous levers      | 2026-2030 | Acting on multiple levers<br>strengthens emissions reduction | <b>TEMPERATURE<br/>IN 2100 : +2.3°C</b>    |



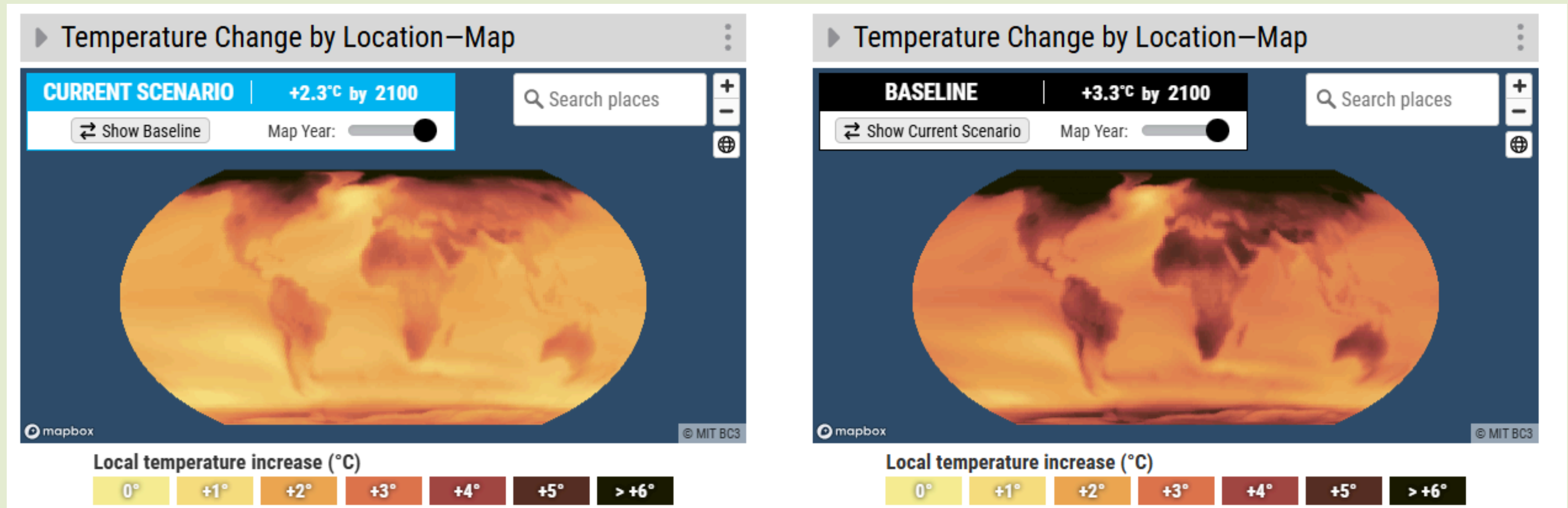
# V. ANALYSIS OF KEY GRAPHS



Map of Temperature Increase in 2050



# V. ANALYSIS OF KEY GRAPHS



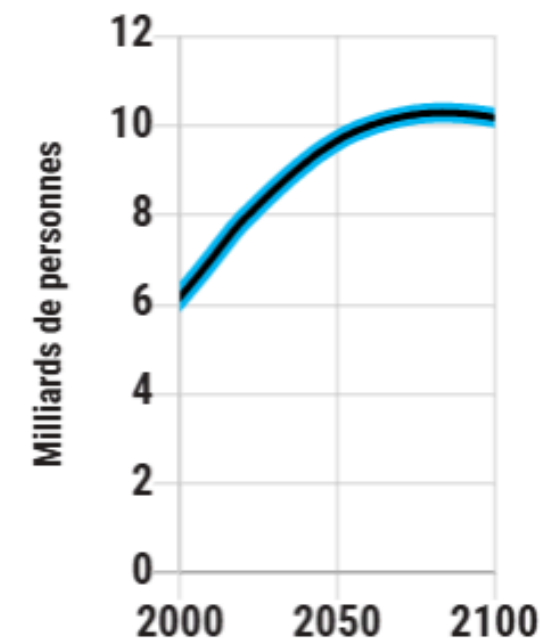
Map of Temperature Increase in 2100

# V. ANALYSIS OF KEY GRAPHS

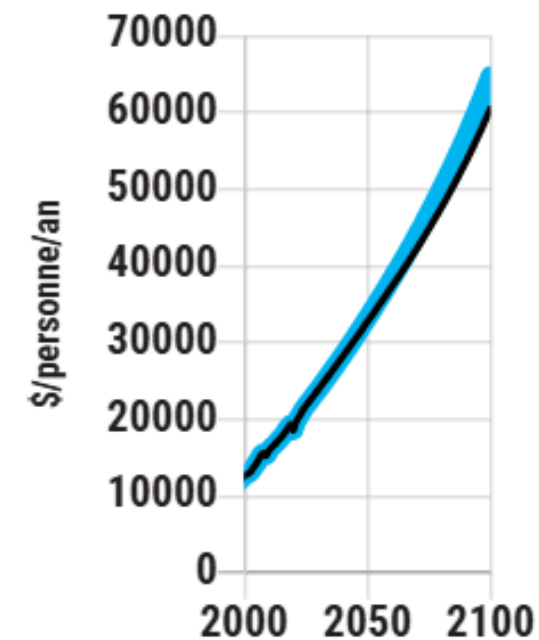
$$CO_2 = P \times \frac{GDP}{P} \times \frac{E}{GDP} \times \frac{CO_2}{E}$$

► Graphiques de Kaya

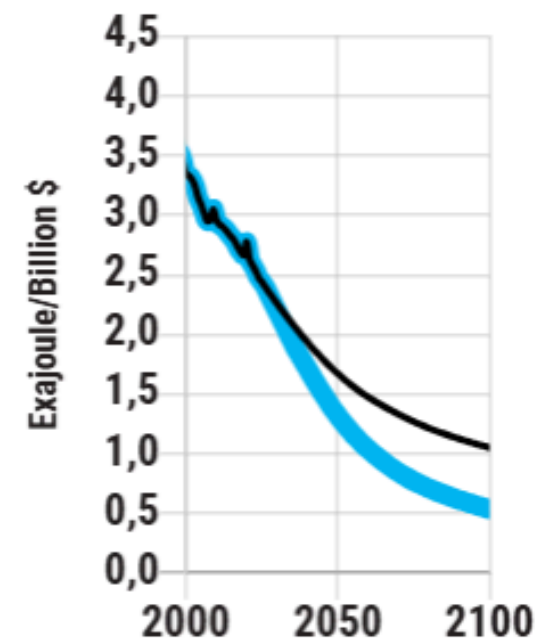
Population mondiale



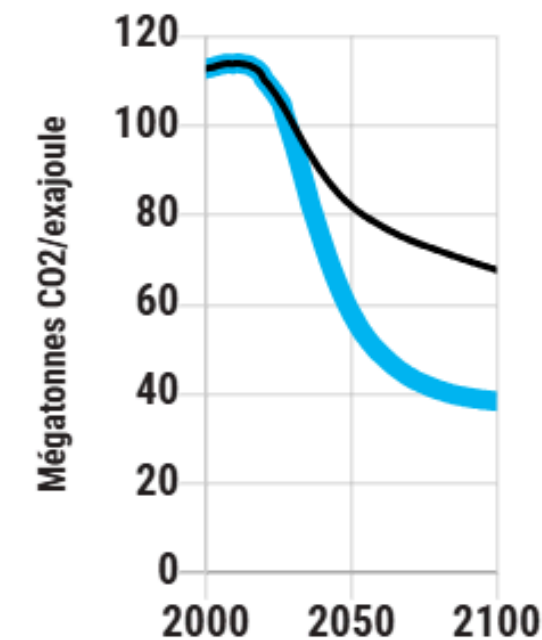
PIB par personne



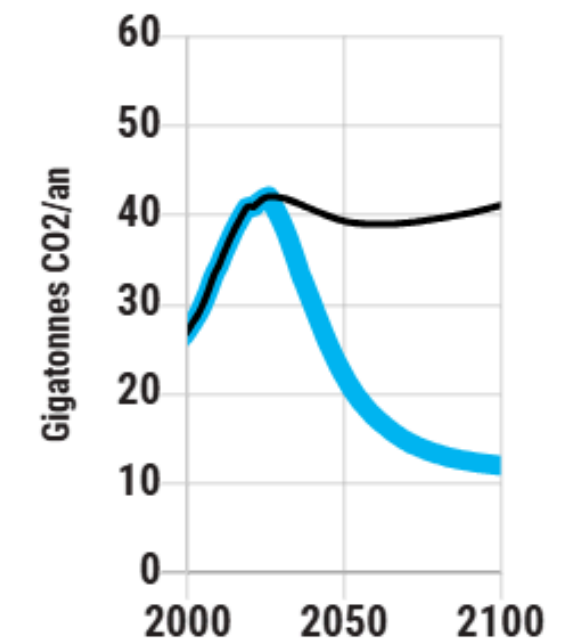
Intensité énergétique du PIB



Intensité carbone de l'énergie finale



Émissions de CO<sub>2</sub> (combustion d'énergie)

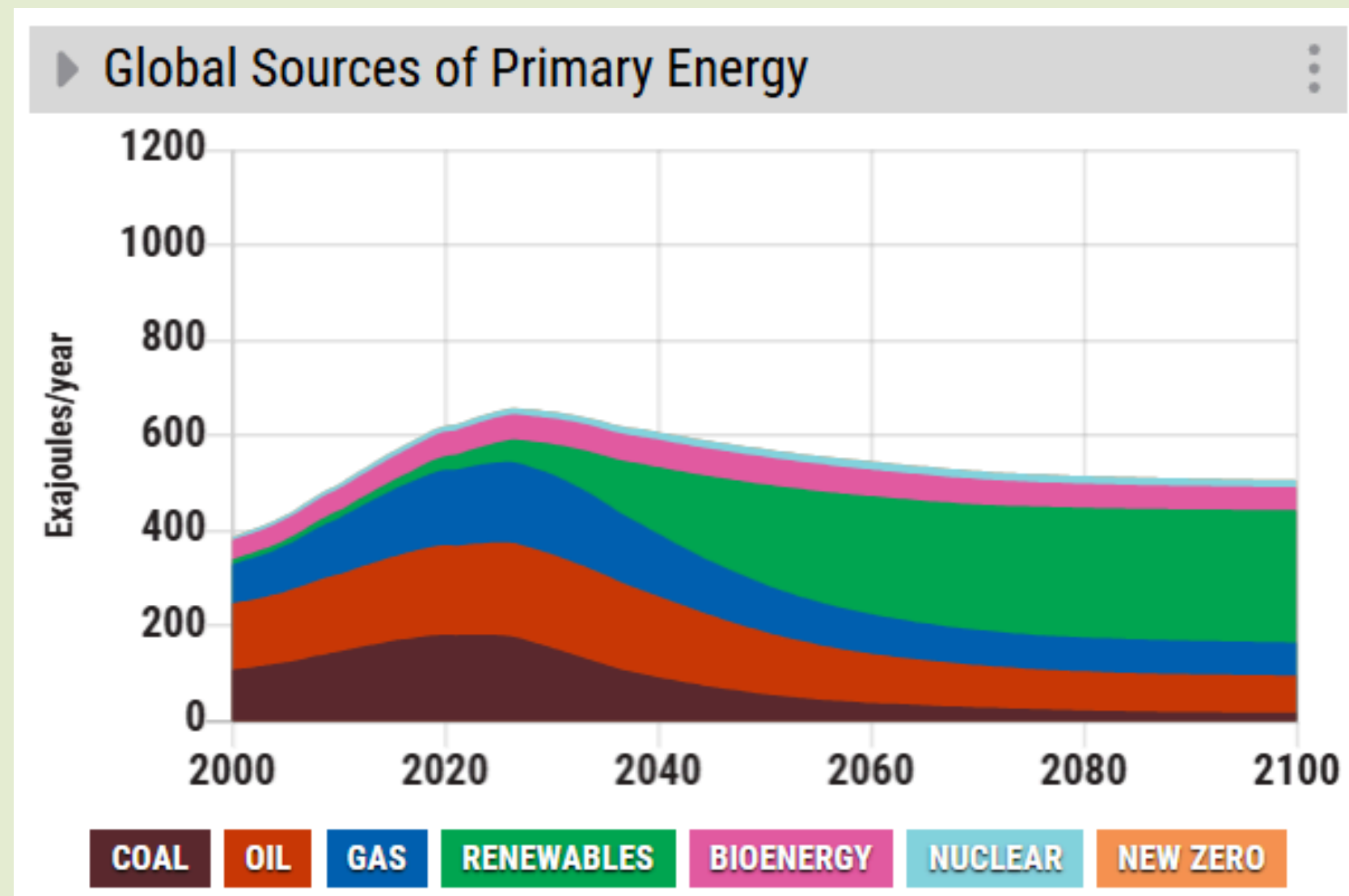


SCÉNARIO DE RÉFÉRENCE

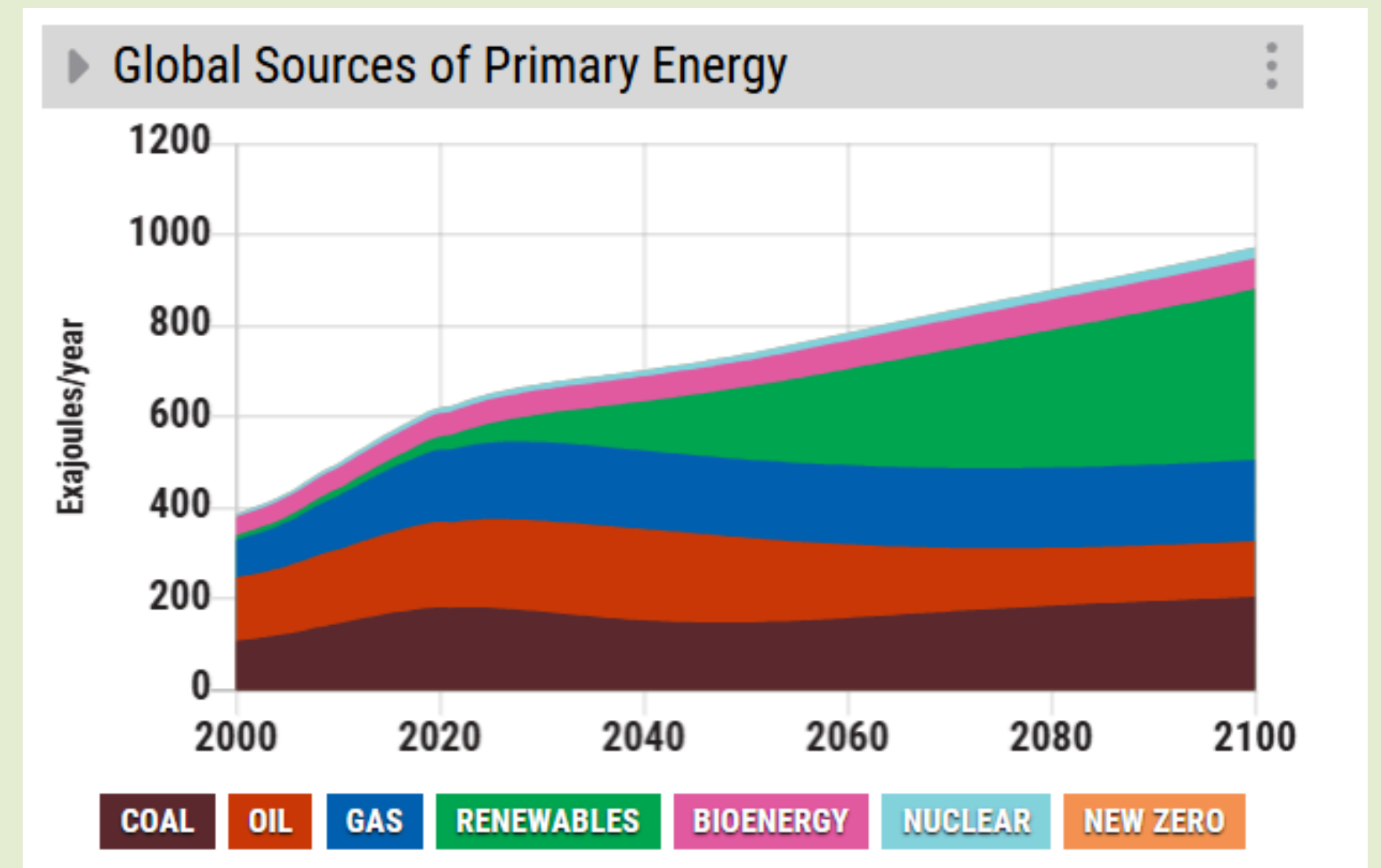
SCÉNARIO ACTUEL

Kaya Graph

# V. ANALYSIS OF KEY GRAPHS



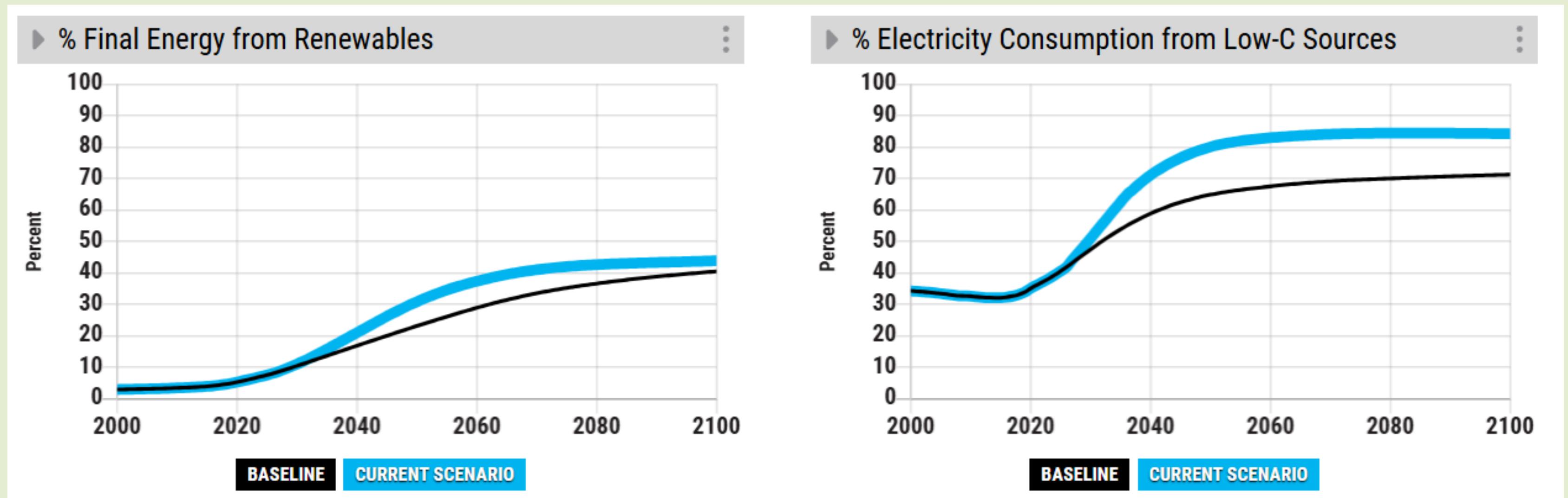
Complete scenario



Reference scenario

Global Primary Energy Sources

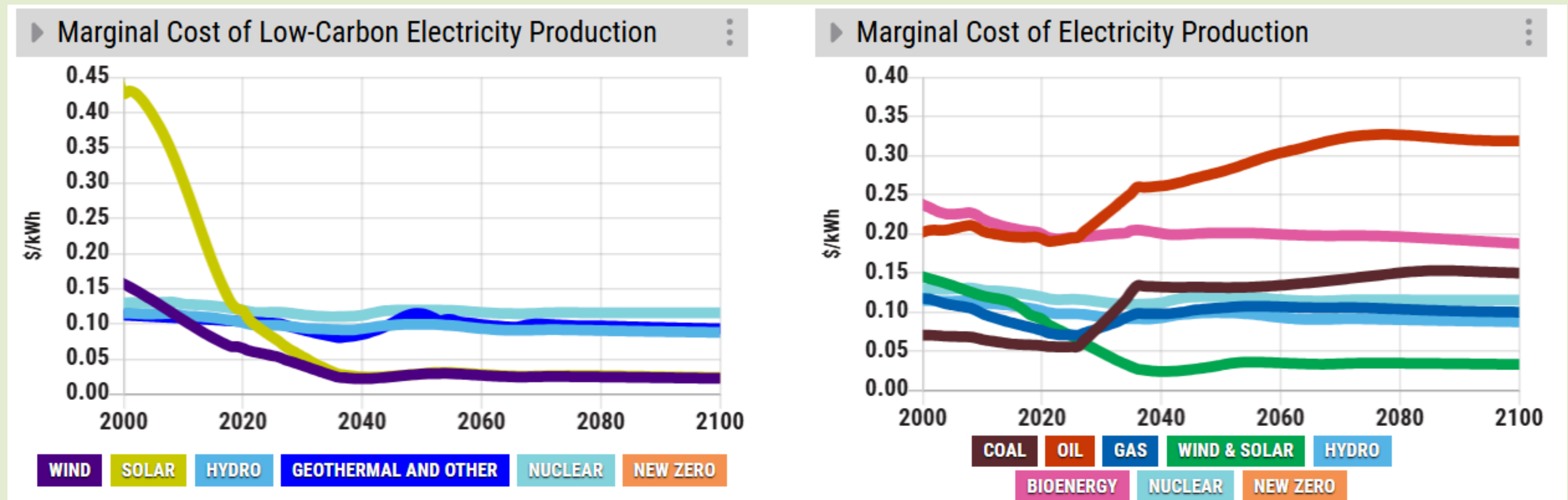
# V. ANALYSIS OF KEY GRAPHS



Renewables Share in Final Energy and Low-Carbon Share in Electricity Consumption (%)



# V. ANALYSIS OF KEY GRAPHS



Marginal Costs of Electricity Production

## **VI. PARALLEL WITH REALITY**

To aim for +1.5°C, even greater efforts would be required:

- A higher and earlier carbon tax
- An energy efficiency rate greater than +2.5%/year
- Even faster electrification of uses
- Massive deployment of renewables coupled with technological advances in storage
- A more ambitious reduction in deforestation than -50%/year



## *VII. LIMITS OF THE MODEL*

- A techno-centric approach neglecting social factors
- Absence of budgetary constraint
- A global model without regional resolution
- Limitations related to land use
- Policy non-linearity and long timeframes

# CONCLUSION

- No single climate lever is sufficient to meet the Paris Agreement targets
- Combined actions (carbon tax, energy efficiency, electrification, reforestation, methane reduction) are necessary
- Our full scenario reaches +2.3°C in 2100, compared to +3.3°C without action
- Synergies between climate policies are essential for an effective transition.
- Solutions exist: the main challenge is political, economic, and social.



***THANK YOU  
FOR YOUR  
ATTENTION***